problem_3

(a) check whethe $\pi = (\frac{4}{3}, \frac{1}{3})$ is an ops to the dual

① feasible ?

Dual: $\min\ 4\pi_1 + 3\pi_2$

s.t. $5\pi_1 - 2\pi_2 \geq 4$

$-\pi_1 + \pi_2 \geq -1$

$2\pi_1 + \pi_2 \geq 3$

$\pi_1, \pi_2 \geq 0$

Check:

$5 \times \frac{4}{3} - 2 \times \frac{1}{3} = 6 \geq 4$ ✓

$-\frac{4}{3} + \frac{1}{3} = -1 \geq -1$ ✓

$2 \times \frac{4}{3} + \frac{1}{3} = 3 \geq 3$ ✓

$\frac{4}{3} \geq 0,\ \frac{1}{3} \geq 0$ ✓

All constraints are satisfied $\Rightarrow$ π is a feasible solution to D

② optimal ?

Let $\begin{cases} \pi_1 (a_1^T x - b_1) = 0 \\ \pi_2 (a_2^T x - b_2) = 0 \\ (c_1 - \pi^T A_1) x_1 = 0 \\ (c_2 - \pi^T A_2) x_2 = 0 \\ (c_3 - \pi^T A_3) x_3 = 0 \end{cases}$ that is $\begin{cases} \frac{4}{3}(5x_1 - x_2 + 2x_3 - 4) = 0 \\ \frac{1}{3}(-2x_1 + x_2 + x_3 - 3) = 0 \\ (4 - 5 \times \frac{4}{3} - 2 \times \frac{1}{3}) x_1 = 0 \\ (-1 - (-\frac{4}{3}) - \frac{1}{3}) x_2 = 0 \\ (3 - 2 \times \frac{4}{3} - \frac{1}{3}) x_3 = 0 \end{cases}$

$\Rightarrow x = (0, \frac{2}{3}, \frac{7}{3})^T$

Check: $5x_1 - x_2 + 2x_3 = 4 \leq 4$ ✓

$-2x_1 + x_2 + x_3 = 3 \leq 3$ ✓

$0 \geq 0,\ \frac{2}{3} \geq 0,\ \frac{7}{3} \geq 0$ ✓

$\Rightarrow x = (0, \frac{2}{3}, \frac{7}{3})^T$ is a feasible solution to P.

Also, we have π^T as a feasible solution to D,

and $\begin{cases} \pi_i (a_i^T x - b_i) = 0, \ \forall i \\ (c_j - \pi^T A_j) x_j = 0, \ \forall j. \end{cases}$

By "Complementary Slackness", $x = (0, \frac{2}{3}, \frac{7}{3})^T$, $\pi = (\frac{4}{3}, \frac{1}{3})$ are optimal solutions to P and D respectively

(b) Find an optimal solution of P: $x = (0, \frac{2}{3}, \frac{7}{3})^T$